

# Buckling of circular cylindrical shells under in-plane and/or pressure loads

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## Abstract

This work presents a semi-analytical approach to calculate rapidly but accurately the buckling onset of metallic and composite circular cylindrical shells with various boundary conditions under in-plane and/or pressure loads by the Rayleigh-Ritz method. Results are compared with analytical solutions and detailed finite element models reported in the literature. The proposed approach allows a quick buckling analysis of circular cylindrical shells, which makes it an ideal candidate to be used as part of an optimization scheme and/or to reduce potentially the number of detailed finite element models employed in the early design phases.

## OPEN ACCESS

**Published:** 30/04/2025

**Accepted:** 05/12/2023

**Submitted:** 31/05/2023

**DOI:** 10.23967/r.matcomp.2025.08.09

## **Keywords:**

Analytical method  
Rayleigh-Ritz  
buckling  
metallic  
composite  
cylinders  
FEM  
método analítico  
pandeo  
metálico  
compuesto  
cilindros

## 1. Introduction

Circular cylindrical shells are extensively used in the aerospace industry. They can be made of metallic or composite materials and designed with or without stiffeners as well as step thickness variations. A typical example of a cylindrical shell could be an aircraft fuselage or rocket pressure fuel tank.

Although the nature of shell buckling is a highly non-linear phenomenon, mostly due to its sensitivity to imperfections, it has been addressed by using a combination of empirical knockdown factors and linear buckling analysis [1]. Several shell theories [1-7] have been developed throughout the years to perform shell buckling analysis. A good overview can be found in [2]. One of the most extensively used is the classical or thin shell theory attributed to Love [3]. Examples of thin shell theories that build upon Love's postulates are those attributed to Timoshenko [4], Flügge [5], Vlasov [6], and Donnell [7] among others.

In the early stages of shell design, when a large number of optimization studies and trends need to be performed, the use of non-linear or even linear detailed finite elements is not feasible due to their complexity and computational cost [1]. An alternative to the use of finite elements is the use of variational or energy methods such as the Rayleigh-Ritz method (e.g., see Ref. [8]). Energy-based methods have been used by many authors (e.g., see Refs. [1,9-10]) and have demonstrated to provide very accurate results. Furthermore, they are proven to be computationally more efficient than traditional finite element models used for structural calculations [11].

This paper focuses on the linear buckling analysis and presents a semi-analytical approach to calculate rapidly but accurately the buckling onset of metallic and composite cylindrical shells with various boundary conditions under in-plane and/or pressure loads by the Rayleigh-Ritz method. The results of the linear buckling analysis are compared with analytical solutions and detailed finite element models reported in the literature.

## 2. Geometry, loads and boundary conditions

The cylindrical shell is circular, of radius ( $R$ ) and length ( $L$ ) and has a thickness ( $t$ ) reinforced with longitudinal stiffeners or stringers and transverse stiffeners or rings. The pitch between the longitudinal and transverse stiffeners is  $d_s$  and  $d_r$ , respectively. An orthogonal curvilinear coordinate system is defined such that  $x$  is the longitudinal direction,  $y$  is the transverse direction following the shell contour and  $z$  is the vertical direction being positive pointing towards the shell exterior. A sketch of the shell under study is shown in [Figure 1](#) below.

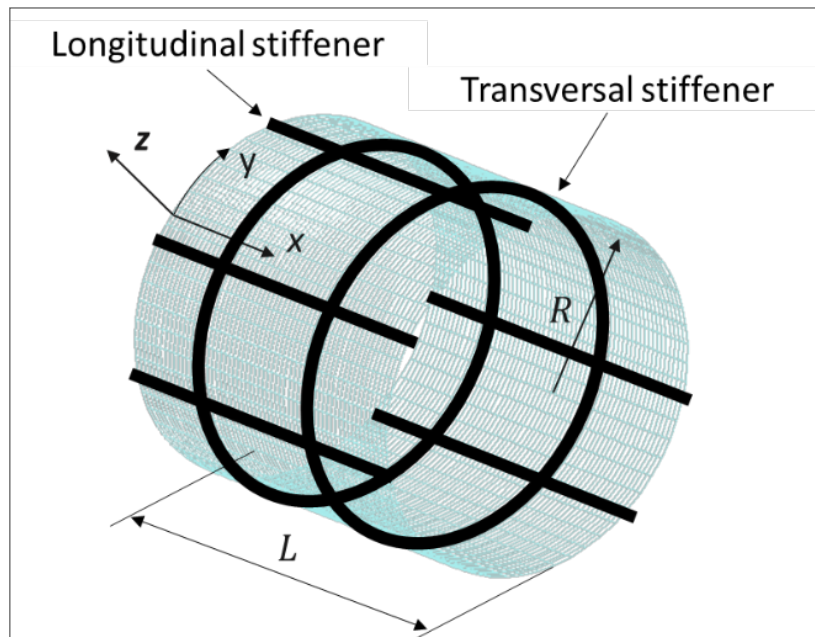


Figure 1. Circular cylindrical shell geometry

The longitudinal and transverse stiffeners are idealised as beam elements and defined by their area ( $A_i$ ), inertia at the centroid ( $I_i$ ), torsional constant ( $J_i$ ), and offset ( $e_i$ ) with respect to the shell midplane where the subscript ( $i$ ) is equal to  $s$  or  $r$ , if the stiffener is longitudinal or transverse, respectively. See [Figure 2](#) below.

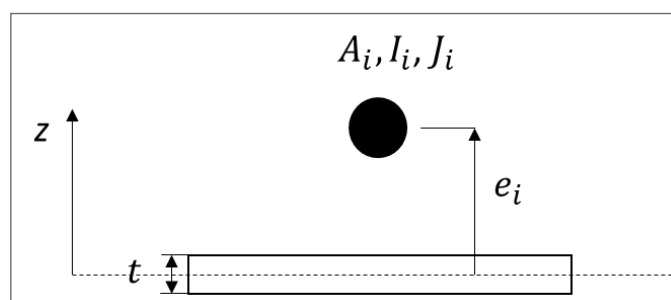


Figure 2. Longitudinal and transverse stiffeners idealisation

The shell is assumed to be under in-plane and/or external pressure loads. For pure external pressure loads  $N_y = -\frac{q_{ext}}{R}$ , similarly, for uniform hydrostatic pressure,  $N_x = -\frac{q_{ext}}{2R}$  and  $N_y = -\frac{q_{ext}}{R}$ , where  $q_{ext}$  is the pressure value. [Figure 3](#) below shows the applied loads.

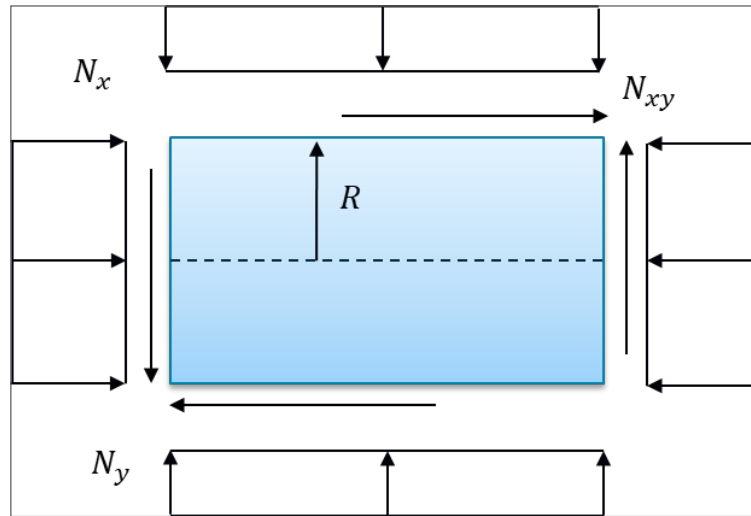


Figure 3. Applied loads

The shell is assumed to be simply supported or clamped at the circular loading edges.

### 3. Fundamental equations of shell theory

The following assumptions following Love's first approximation [3] are considered:

- The shell is thin. That is the thickness of the shell is small when compared to other dimensions.
- Displacement, strains and rotations are small.
- Kirchhoff's hypotheses. That is, the normals to the middle surface remain straight before and after deformation and suffer no extension. In other words, transverse shear and through thickness strains are neglected ( $\gamma_{xz} = \gamma_{yz} = \epsilon_z = 0$ ).
- The through thickness stress is small and can be neglected ( $\sigma_z = 0$ ).
- Material behaviour is linearly elastic. Materials can be metallic or composites.

The relations between the in-plane strains, curvatures and displacements are linear with  $z$  and are obtained using Donnell's shell theory [7]. Thus,

$$\{\epsilon\} = \{\epsilon^0\} + z \{\kappa\} \quad (1)$$

where,  $\epsilon = \{\epsilon_x, \epsilon_y, \epsilon_{xy}\}^T$  is the vector of strains,  $\epsilon^0 = \{\epsilon_x^0, \epsilon_y^0, \epsilon_{xy}^0\}^T$  is the vector of the in-plane strains,  $\kappa = \{\kappa_x, \kappa_y, \kappa_{xy}\}^T$  is the vector of the curvatures, with

$$\begin{aligned} \epsilon_x^0 &= \frac{\partial u_0}{\partial x} & \epsilon_y^0 &= \frac{\partial v_0}{\partial y} + \frac{w_0}{R} & \epsilon_{xy}^0 &= \frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x} \\ \kappa_x &= -\frac{\partial^2 w_0}{\partial x^2} & \kappa_y &= -\frac{\partial^2 w_0}{\partial y^2} & \kappa_{xy} &= -2\frac{\partial^2 w_0}{\partial x \partial y} \end{aligned} \quad (2)$$

where  $u_0$ ,  $v_0$  and  $w_0$  are the displacements in  $x$ ,  $y$  and  $z$  directions, respectively.

The material constitutive equations of the shell are obtained by (e.g. [8]),

$$\begin{Bmatrix} N \\ M \end{Bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix} \begin{Bmatrix} \epsilon^0 \\ \kappa \end{Bmatrix} \quad (3)$$

where  $N = \{N_x, N_y, N_{xy}\}^T$  is the vector of running loads,  $M = \{M_x, M_y, M_{xy}\}^T$  is the vector of running moments,  $A$  is the membrane stiffness matrix,  $B$  is the membrane-bending coupling stiffness matrix and  $D$  is the bending stiffness matrix.

The membrane, membrane-bending and bending stiffness matrix terms of the shell considering the longitudinal and transverse stiffener properties are as follows [1];

$$\begin{aligned}
 \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{12} & A_{22} & A_{23} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} &= \begin{bmatrix} A_{11}^{shell} & A_{12}^{shell} & 0 \\ A_{12}^{shell} & A_{22}^{shell} & 0 \\ 0 & 0 & A_{33}^{shell} \end{bmatrix} + \begin{bmatrix} \frac{E_s A_s}{d_s} & 0 & 0 \\ 0 & \frac{E_r A_r}{d_r} & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
 \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{12} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33} \end{bmatrix} &= \begin{bmatrix} B_{11}^{shell} & B_{12}^{shell} & 0 \\ B_{12}^{shell} & B_{22}^{shell} & 0 \\ 0 & 0 & B_{33}^{shell} \end{bmatrix} + \begin{bmatrix} \frac{E_s A_s}{d_s} e_s & 0 & 0 \\ 0 & \frac{E_r A_r}{d_r} e_r & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
 \begin{bmatrix} D_{11} & D_{12} & D_{13} \\ D_{12} & D_{22} & D_{23} \\ D_{13} & D_{23} & D_{33} \end{bmatrix} &= \begin{bmatrix} D_{11}^{shell} & D_{12}^{shell} & 0 \\ D_{12}^{shell} & D_{22}^{shell} & 0 \\ 0 & 0 & D_{33}^{shell} \end{bmatrix} + \\
 &\begin{bmatrix} \frac{E_s}{d_s} (I_s + A_s e_s^2) & 0 & 0 \\ 0 & \frac{E_r}{d_r} (I_r + A_r e_r^2) & 0 \\ 0 & 0 & \frac{1}{4} \left( \frac{G_s J_s}{d_s} + \frac{G_r J_r}{d_r} \right) \end{bmatrix}
 \end{aligned} \tag{4}$$

where  $E_i$  and  $G_i$  ( $i = s$  or  $r$ ) are the Young's and Shear moduli for the longitudinal and transverse stiffeners, respectively. For this work, shell anisotropy is not considered.

## 4. Problem formulation

The Rayleigh-Ritz method is used to perform the buckling analysis of the circular cylindrical shell. This is an approximate method that uses energy principles that are equivalent to the governing differential equations (i.e., Euler-Lagrange) and natural boundary conditions [12]. The energy principle used in this work is the principle of minimum potential energy (e.g., see Ref. [8]).

In the Rayleigh-Ritz method the displacements are approximated in the form of a finite double series, such that

$$\begin{aligned} u_0(x,y) &= \sum_i^I \sum_j^J u_{ij} X_{i_x}(x) Y_j(y) \\ v_0(x,y) &= \sum_i^I \sum_j^J v_{ij} X_i(x) Y_{j_y}(y) \\ w_0(x,y) &= \sum_i^I \sum_j^J w_{ij} X_i(x) Y_j(y) \end{aligned} \quad (5)$$

where the functions  $X_i(x)$  and  $Y_j(y)$  are the  $i$ th and  $j$ th known functions in the  $x$  and  $y$  directions, respectively that must satisfy the geometrical boundary conditions in the whole domain [13].  $u_{ij}$ ,  $v_{ij}$  and  $w_{ij}$  are unknown constants for the displacements in  $x$ ,  $y$  and  $z$ , respectively and  $I$  and  $J$  are the number of terms in the series in the  $x$  and  $y$  direction, respectively. Note that  $(\cdot)_{,\alpha}$  denotes  $\frac{\partial}{\partial \alpha}(\cdot)$ .

The geometrical boundary conditions considered for the shell at the circular edges (e.g., see Ref. [14]) and are shown in Table 1 below.

| Type   | Geometrical boundary condition                                                                             |
|--------|------------------------------------------------------------------------------------------------------------|
| S2(SS) | $v(0,y) = v(L,y) = 0$<br>$w(0,y) = w(L,y) = 0$<br>$w_{,x}(0,y) = w_{,x}(L,y) = 0$                          |
| C1(CC) | $u(0,y) = u(L,y) = 0$<br>$v(0,y) = v(L,y) = 0$<br>$w(0,y) = w(L,y) = 0$<br>$w_{,x}(0,y) = w_{,x}(L,y) = 0$ |

Table 1. Geometrical boundary conditions at the circular edges

The trigonometric functions for in the  $x$  direction that satisfy the simply supported and clamped conditions (e.g., see Ref. [15]) in Table 1 are given by,

$$\begin{aligned} X_i(x) &= \sin\left(\frac{i\pi x}{L}\right) \quad i = 1, \dots, I \quad \rightarrow SS \\ X_i(x) &= \cos\left((i-1)\frac{\pi x}{L}\right) - \cos\left((i+1)\frac{\pi x}{L}\right) \quad i = 1, \dots, I \quad \rightarrow CC \end{aligned} \quad (6)$$

The trigonometric function in the  $y$  direction is given by

$$Y_j(y) = \sin\left(\frac{jy}{R}\right) \quad j = 1, \dots, J \quad (7)$$

For the buckling problem, the principle of minimum potential energy takes the form,

$$\partial V = \delta(U + \lambda W) = 0 \quad (8)$$

with

$$\begin{aligned} U &= \frac{1}{2} \iiint (\epsilon_x \sigma_x + \epsilon_y \sigma_y + \gamma_{xy} \tau_{xy}) dV \\ W &= -\frac{1}{2} \iint \left[ N_x \left( \frac{\partial w_0}{\partial x} \right)^2 + N_y \left( \frac{\partial w_0}{\partial y} \right)^2 + 2N_{xy} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial y} \right] dA \end{aligned} \quad (9)$$

where  $V$  is the potential energy of the shell,  $U$  is the strain energy of the shell including the longitudinal and transverse stiffeners,  $W$  is work of external forces and  $\lambda$  is the critical buckling load factor.

The expression of the potential energy of the shell is found by substituting Equations (1)-(4) in Equation (9) and rearranging,

$$\begin{aligned}
 V = & \frac{1}{2} \iint \left[ A_{11} \left( \frac{\partial u_0}{\partial x} \right)^2 + 2A_{12} \frac{\partial u_0}{\partial x} \left( \frac{\partial v_0}{\partial y} + \frac{w_0}{R} \right) + A_{22} \left( \frac{\partial v_0}{\partial y} + \frac{w_0}{R} \right)^2 + \right. \\
 & A_{33} \left( \frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x} \right)^2 \Big] dA - \iint \left[ B_{11} \frac{\partial u_0}{\partial x} \frac{\partial^2 w_0}{\partial x^2} + 2B_{12} \left( \left( \frac{\partial v_0}{\partial y} + \frac{w_0}{R} \right) \frac{\partial^2 w_0}{\partial x^2} + \frac{\partial u_0}{\partial x} \frac{\partial^2 w_0}{\partial y^2} \right) + B_{22} \left( \frac{\partial v_0}{\partial y} + \frac{w_0}{R} \right) \frac{\partial^2 w_0}{\partial xy^2} + 4B_{33} \left( \frac{\partial u_0}{\partial y} + \right. \right. \\
 & \left. \left. \frac{\partial v_0}{\partial x} \right) \frac{\partial^2 w_0}{\partial x \partial y} \right] dA + \frac{1}{2} \iint \left[ D_{11} \left( \frac{\partial^2 w_0}{\partial x^2} \right)^2 + 2D_{12} \frac{\partial^2 w_0}{\partial x^2} \frac{\partial^2 w_0}{\partial y^2} + \right. \\
 & D_{22} \left( \frac{\partial^2 w_0}{\partial y^2} \right)^2 + 4D_{33} \left( \frac{\partial^2 w_0}{\partial x \partial y} \right)^2 \Big] dA - \lambda \frac{1}{2} \iint \left[ N_x \left( \frac{\partial w_0}{\partial x} \right)^2 + \right. \\
 & \left. N_y \left( \frac{\partial w_0}{\partial y} \right)^2 + 2N_{xy} \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial y} \right] dA
 \end{aligned} \tag{10}$$

The solution of the problem is found by substituting Equation (5) in Equation (10) and minimising with respect to each of the displacements' unknown coefficients. Hence,

$$\begin{aligned}
 \frac{\partial V}{\partial u_{ij}} = 0 \quad \frac{\partial V}{\partial v_{ij}} = 0 \quad \frac{\partial V}{\partial w_{ij}} = 0 \\
 i = 1, \dots, I \text{ and } j = 1, \dots, J
 \end{aligned} \tag{11}$$

This generates a system of  $3 \cdot I \cdot J$  linear equations with  $3 \cdot I \cdot J$  unknowns. Thus,

$$\left( \begin{bmatrix} K_{uu} & K_{uv} & K_{uw} \\ K_{uv}^T & K_{vv} & K_{vw} \\ K_{uw}^T & K_{vw}^T & K_{ww} \end{bmatrix} + \lambda \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & L_{ww} \end{bmatrix} \right) \cdot \begin{Bmatrix} u \\ v \\ w \end{Bmatrix} = 0 \tag{12}$$

where  $\mathbf{u}$ ,  $\mathbf{v}$  and  $\mathbf{w}$  are the unknown coefficients of the displacement vectors, each of  $I \cdot J$  dimension,  $\mathbf{K}_{mn}$  ( $m, n = u, v, w$ ) are squared matrices each of  $I \cdot J \times I \cdot J$  dimensions related to the shell's stiffnesses and  $\mathbf{L}_{ww}$  is a squared matrix of  $I \cdot J \times I \cdot J$  dimensions, related to the applied loads. This linear system is homogeneous and can be solved as an eigenvalue problem. The first eigenvalue will provide the critical buckling load such that,

$$\begin{Bmatrix} N_x^{cr} \\ N_y^{cr} \\ N_{xy}^{cr} \end{Bmatrix} = \lambda \begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix} \quad (13)$$

## 5. Numerical results

The semi-analytical method presented in this paper was developed in Fortran [16]. Integrals to obtain the matrices in Equation (12) were computed numerically. Results from the semi-analytical approach were compared against analytical solutions and detailed finite element models reported in the literature. Sensitivity studies were performed to determine the number of terms in the  $x$  and  $y$  directions in the series that guarantee accurate predictions for the eigenvalue obtained with Equation (12). Even for a large number of terms in the series the method was very fast providing result within seconds. Unstiffened and stiffened circular cylindrical shells made of metallic or composite materials subjected to different states of loading were considered.

First, results and comparisons are presented for unstiffened shells. Table 2 shows the non-dimensional buckling pressure  $\frac{q_{ext}^{cr} R L^2}{\pi^2 D}$  for isotropic unstiffened circular cylinders with simply supported edges and subjected to uniform external pressure for several Batdorf parameters ( $Z$ ). The shell is made of a metallic material with  $\nu = 0.3$  (Poisson's ratio). From Table 2, it can clearly be seen that the present method provides identical and nearly identical results to those reported in [1] and [17], respectively. The

maximum difference is approximately 0.6% for  $Z = 100000$ .

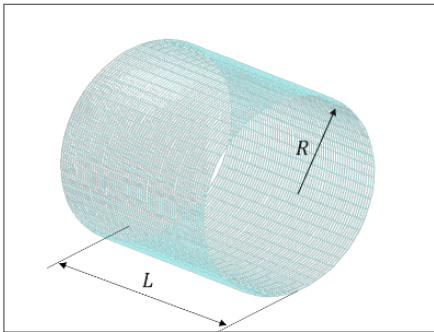
| Unstiffened shell <sup>a</sup>                                                                                                                                                                                                                                                                   |  | Reisen                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------|
|                                                                                                                                                                                                                |  | $\frac{4\pi^2}{Rt} \sqrt{1 + \frac{L^2}{4\pi^2 R^2}}$ |
|                                                                                                                                                                                                                                                                                                  |  | Donnell                                               |
|                                                                                                                                                                                                                                                                                                  |  | 50.836                                                |
|                                                                                                                                                                                                                                                                                                  |  | (1,44)[10]                                            |
|                                                                                                                                                                                                                                                                                                  |  | 10.884                                                |
|                                                                                                                                                                                                                                                                                                  |  | (1,26)[10]                                            |
|                                                                                                                                                                                                                                                                                                  |  | 30.054                                                |
| $= \frac{Et^3}{12(1 - \nu^2)}$ <p><sup>b</sup> Numbers in parentheses are the number of longitudinal half-waves and circumferential waves, respectively</p> <p><sup>c</sup> Numbers in brackets are the number of terms in the trigonometric series for the x and y directions, respectively</p> |  | (1,15)[10]                                            |
|                                                                                                                                                                                                                                                                                                  |  | 106.95                                                |
|                                                                                                                                                                                                                                                                                                  |  | (1,9)[10]                                             |
|                                                                                                                                                                                                                                                                                                  |  | 380.80                                                |
|                                                                                                                                                                                                                                                                                                  |  | (1,5)[10]                                             |

Table 2. Non-dimensional buckling factor – isotropic unstiffened circular cylinders with simply supported edges and under external pressure.

Number in parentheses are the number of terms used in the trigonometric series in the x and y directions.

It can also be seen that, the number of longitudinal half-waves and circumferential waves obtained with the present method coincided with those reported in [1]. Figure 4 displays two examples of the buckling modes obtained with the present method that are reported in Table 2.

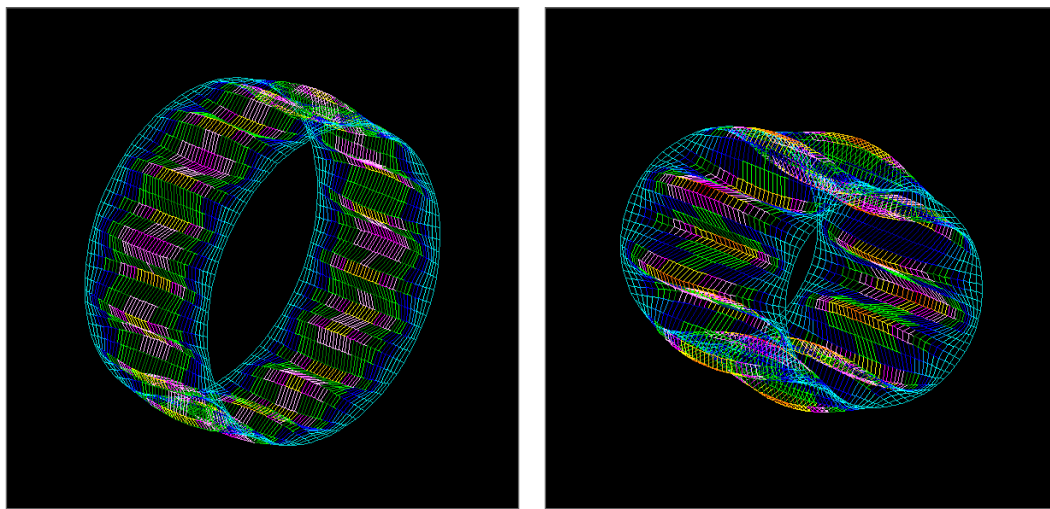


Figure 4. Examples of buckling modes for isotropic unstiffened circular cylinders with simply supported edges and under external pressure for  $Z=1000$  (left) and  $Z =10000$  (right).

Next, Table 3 below provides the buckling pressure ( $\times 10^{-4}MPa$ ) for isotropic unstiffened circular cylinders with simply supported edges and subjected to external pressure. The shell is made of a metallic material with  $E = 200GPa$  and  $\nu = 0.3$ . Results are provided using  $10 \times 20$  and  $10 \times 30$  terms in the trigonometric series for  $\frac{R}{t} = 300$  and  $\frac{R}{t} = 3000$ , respectively. From Table 3, it is clearly seen that the present method provides identical results to those reported in [1] and compares well with the detailed finite element models reported in [18]. The maximum difference in this case is approximately 5.2% for  $\frac{R}{t} = 300$  and  $\frac{R}{t} = 0.5$ .

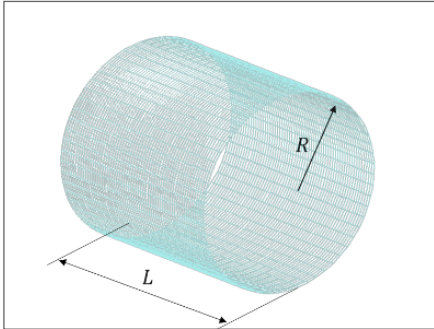
| Unstiffened shell                                                                  |  | Ref    |
|------------------------------------------------------------------------------------|--|--------|
|  |  | [48]   |
|                                                                                    |  | DEM    |
|                                                                                    |  | 0002.0 |
|                                                                                    |  | 1233.6 |
|                                                                                    |  | 012.8  |
|                                                                                    |  | 412.02 |
|                                                                                    |  | 239.83 |
|                                                                                    |  | 0.842  |
|                                                                                    |  | 3.809  |
|                                                                                    |  | 2.889  |
|                                                                                    |  | 3.256  |
|                                                                                    |  | 0.748  |

Table 3. Buckling pressure – isotropic unstiffened circular cylinders with simply supported edges and under external pressure.

Shells with different boundary conditions were also considered. Table 4 below provides the buckling pressure ( $\times 10^3 Pa$ ) for isotropic unstiffened circular cylinders with clamped edges and subjected to external pressure. As before, the shell is made of a metallic material with  $E = 200 GPa$  and  $\nu = 0.3$ . Results are provided using 35x20 and 35x35 terms in the trigonometric series for  $\frac{R}{t} = 300$  and  $\frac{R}{t} = 3000$ , respectively. The results in Table 4 indicate that the present method compares well with the detailed finite element models considering clamped edges reported in [18]. The maximum difference observed is approximately 4.7% for  $\frac{R}{t} = 300$  and  $\frac{L}{R} = 0.5$ .

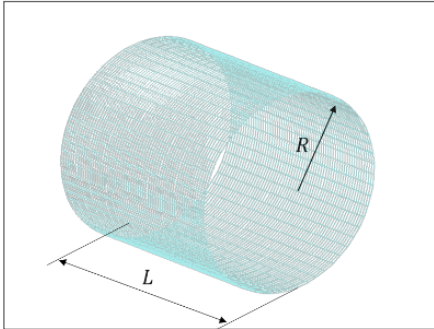
| Unstiffened shell                                                                  |  | Present |
|------------------------------------------------------------------------------------|--|---------|
|  |  | [18]    |
|                                                                                    |  | FEM     |
|                                                                                    |  | 0.002   |
|                                                                                    |  | 1703    |
|                                                                                    |  | 880.4   |
|                                                                                    |  | 383.5   |
|                                                                                    |  | 352.4   |
|                                                                                    |  | 0.0006  |
|                                                                                    |  | 5.520   |
|                                                                                    |  | 2.799   |
|                                                                                    |  | 3.804   |
|                                                                                    |  | 5.128   |

Table 4. Buckling pressure – isotropic unstiffened circular cylinders with clamped edges under external pressure

In addition, shells made of composite materials were also studied. Table 5 below provides non-dimensional buckling loads  $\frac{N_x^{cr} L^2}{t^3 E_2}$  for unsymmetric cross-ply circular cylinders with simply supported edges and subjected to axial compression. The radius ( $R$ ) and thickness of the shell ( $t$ ) are 10 and 0.01 inches respectively. Ply properties used are  $E_{11} = 30 \times 10^6 \text{psi}$ ,  $E_{22} = 0.75 \times 10^6 \text{psi}$ ,  $G_{12} = 3.75 \times 10^6 \text{psi}$  and  $\nu_{12} = 0.25$ . Results are provided using 10x10 terms in the trigonometric series. From Table 5, it can be observed that the present method provides results that are slightly stiffer than those reported in [19]. The maximum difference in this case is approximately 8.6% for the (0/90) lay-up with  $Z = 100$ .

| Unstiffened shell |  | Rayson                            |
|-------------------|--|-----------------------------------|
|                   |  | $\frac{4.9}{Rt} \sqrt{1 - \nu^2}$ |
|                   |  | Donnell                           |
|                   |  | (0.8503)                          |
|                   |  | (0.8704)                          |
|                   |  | (0.9108)                          |
|                   |  | (0.9210)                          |
|                   |  | (0.9229)                          |
|                   |  | (0.9282)                          |
|                   |  | (0.9309)                          |
|                   |  | (0.936)                           |
|                   |  | (0.9507)                          |
|                   |  | (0.9608)                          |
|                   |  | (0.9783)                          |
|                   |  | (0.9937)                          |

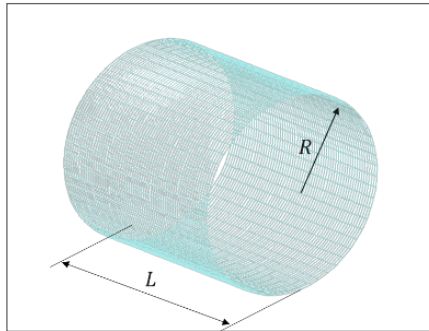


Table 5. Non-dimensional buckling loads – unsymmetric cross-ply unstiffened circular cylinders with simply supported edges and under external pressure

Secondly, results for shells stiffened with longitudinal stiffeners or stringers are presented. Table 6 gives the non-dimensional buckling coefficient  $\frac{2N_x^{cr} R^2}{D}$  for isotropic circular cylinders with eccentric stringers with simply supported edges and subjected to axial. The shell is made of a metallic material with  $\nu = 0.3$ . Results are provided using 10x10 terms in the trigonometric series. Table 6 shows the present method provides identical results to those reported in [1] and in [20] (as cited in [1]).

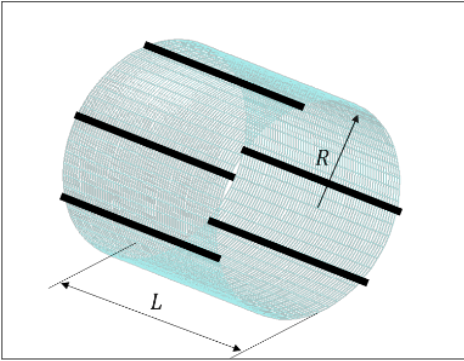
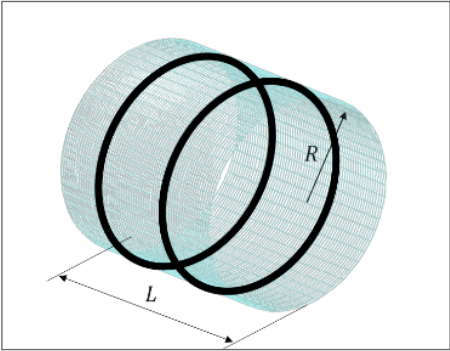
|                                                                                     |                                              |
|-------------------------------------------------------------------------------------|----------------------------------------------|
| <p>Shell stiffened with stringers<sup>a</sup></p>                                   | <p>Ref. [4] as cited in Donnell Ref. [1]</p> |
|  | <p>Donnell 8064 04838 19750</p>              |
| $= \frac{Et^3}{12(1-\nu^2)} ; \frac{I_s}{d_s t^3} = 5; J_s = 0; \frac{e_s}{t} = +5$ | <p>7689 00704 16258 28200 34929 46578</p>    |

Table 6. Non-dimensional buckling coefficient – isotropic circular cylinders with stringers, simply supported edges and under axial load

Finally, results for shells stiffened with transverse stiffeners or rings are presented. Table 7 provides the non-dimensional buckling pressure  $\frac{q_{ext}^{cr} R^3}{D}$  for isotropic circular cylinders stiffened with eccentric rings with simply supported edges and subjected to uniform external pressure. The shell is made of a metallic material with  $\nu = 0.3$ . Results are provided using 10x10

terms in the trigonometric series. From [Table 7](#), it can clearly be seen that the present method provides results that are identical to those reported in [1] and nearly identical to those in [21-22] (as cited in [1]).

|                                                                                     |                                            |
|-------------------------------------------------------------------------------------|--------------------------------------------|
| Shell stiffened with rings <sup>a</sup>                                             |                                            |
|  | Ref. [1]-[22] as cited in Donnell Ref. [1] |
|                                                                                     | Donnell 2043                               |
|                                                                                     | $e_r + 70070$                              |
|                                                                                     | $e_r + 2500,$                              |
|                                                                                     | $e_r + 5883$                               |
|                                                                                     | $e_r - 75028$                              |
|                                                                                     | $e_r - 20208$                              |
|                                                                                     | $e_r -$                                    |

$$= \frac{Et^3}{12(1-\nu^2)}; \frac{A_r}{d_r t^3} = 0.5; \frac{I_r}{d_r t^3} = 5; J_{rs} = 0; \frac{e_r}{t} = \pm 5$$

<sup>b</sup> $e_{r+}$  and  $e_{r-}$  indicate positive and negative ring eccentricity, respectively

Table 7. Non-dimensional buckling pressure – isotropic circular cylinders with rings, simply supported edges and under external pressure

## 6. Concluding remarks

A semi-analytical approach using the Rayleigh-Ritz method was developed to calculate the linear buckling onset of metallic and composite circular cylindrical shells with various boundary conditions and subjected to in-plane and/or pressure loads. Donnell's [7] shell theory was used due to its simplicity. Results for unstiffened and stiffened shells were compared with analytical solutions and detailed finite elements reported in the literature. Overall, the present method provided accurate results. For the metallic shells considered the present method provided results identical to the analytical results reported in [1] and close to the detailed finite element models results found in [18]. The maximum differences with detailed finite elements were 5.2%. For the composite shells considered, the present method provided slightly stiffer results than the analytical ones reported in [19]. For this case, the maximum differences were 8.6%. The present method was computationally fast even when large number of terms were used in the trigonometric series. Both its accuracy and computational speed makes the present method an ideal candidate to be used as part of an optimization scheme and/or to reduce potentially the number of detailed finite element models employed in the early design phases.

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